

HARIRAMAKRRISHNAN RAMACHANDRAN

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PROFESSIONAL EXPERIENCE

Software Engineer — HCL Technologies, Chennai, India

Sep 2021 – Jan 2025

- Designed and implemented **agentic LLM systems** that integrated **external tools** (APIs, databases) for precision task execution, reducing manual intervention by **30%**.
- Adapted **compact language models** to specialized domains using techniques such as **fine-tuning** and **parameter-efficient methods**, improving downstream task performance by an average of **15%**.
- Developed **Python prototypes** for novel AI research, collaborating with Senior Research Scientists to rapidly build and test **Proof-of-Concepts**.
- Authored technical documentation and research summaries on advancements in **LLM domain adaptation** and **knowledge integration**, contributing to internal knowledge sharing.
- Articulated complex research findings into clear, persuasive presentations for both academic and business audiences, translating research outcomes into compelling **value propositions**.
- Implemented custom **neural network layers** and **loss functions** in Python using **PyTorch** for experimental AI models.
- Containerized AI model serving endpoints using **Docker** and deployed to a **Kubernetes** cluster, ensuring scalable and reliable inference.

SKILLS

AI/ML Core – Machine learning, Supervised learning, Embeddings, Evaluation metrics, Data preprocessing, Neural networks, Transformers, LLM domain adaptation, Knowledge integration, Reasoning techniques, Small Language Models (SLMs), Agentic LLMs, Tool Use

Programming – Python, PyTorch, TensorFlow, SQL

Cloud & Containerization – AWS, Docker, Kubernetes

Research & Development – Experimental design, Research synthesis, Prototype development, Research dissemination, Strategic storytelling

Data & Integration – APIs, Databases, Document collections, Knowledge graphs, Rule-based systems

PROJECTS

LLM-Powered Knowledge Integration System

- Developed a system to integrate external knowledge sources (databases, document collections) into LLMs using **Retrieval-Augmented Generation (RAG)**.
- Built a **Python-based agent** that utilized a vector database to retrieve relevant information, enabling LLMs to perform more accurate and verifiable tasks.
- Fine-tuned a compact language model on domain-specific documents, achieving a **20% improvement** in question-answering accuracy for specialized queries.

AI Agent for Automated API Interaction

- Designed and implemented an agentic LLM system capable of interacting with external APIs (e.g., for data retrieval and task execution).
- Integrated LLMs with a Python-based tool execution engine, enabling the system to perform multi-step tasks with high precision and low error rates.
- Developed a framework for evaluating agent performance on complex tasks, tracking metrics such as task completion rate and precision, reaching **~92% success rate** on benchmark tests.

EDUCATION

M.Sc. Data Analytics — National College of Ireland, Dublin

Feb 2026

B.E. Computer Science — SNS College of Technology, Coimbatore, India

2021

CERTIFICATIONS

- IELTS Academic: Overall Band Score **7 out of 9**, issued by IELTS Official, Test Date: 28/AUG/2023
- **Full Stack: Java Spring Boot** and Angular (Great Learning)
- Certificate of Completion in Web Development from **University of California**, Davis (Coursera)
- Certificate of Completion in Python programming from **University of Michigan** (Coursera)